Bryant Farms Road Extension: Pavement Alternatives

EMGT 6930

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Introduction

- The south Charlotte community of Ballantyne has seen recent rapid population growth
- The existing transportation infrastructure can no longer support local needs
- The Charlotte Department of Transportation (CDOT) has allocated funding to public capital transportation projects in the community
- **Project:** Bryant Farms Road Extension Phase II
- **Delivery Method:** Design-Build
- **Project Status:** Active Procurement
- **Project Value:** Initially estimated at $53 million in 2022

Expected Award
Q4 2026

Expected Completion
2029-2030

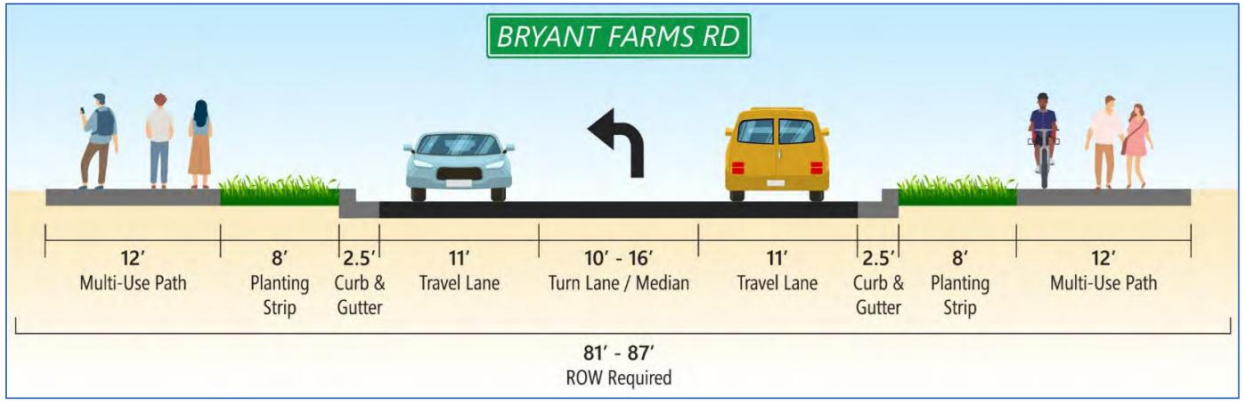
Scope of Work

Design

- Roadway
- Grading / Stormwater
- Landscape

Construction

- 1.2 miles roadway extension
- Shared use paths
- Local restoration



Potential Typical Section Option 2

Figure 1: Bryant Farms Road Section Exhibit (CDOT Bryant Farms RFP 2025)

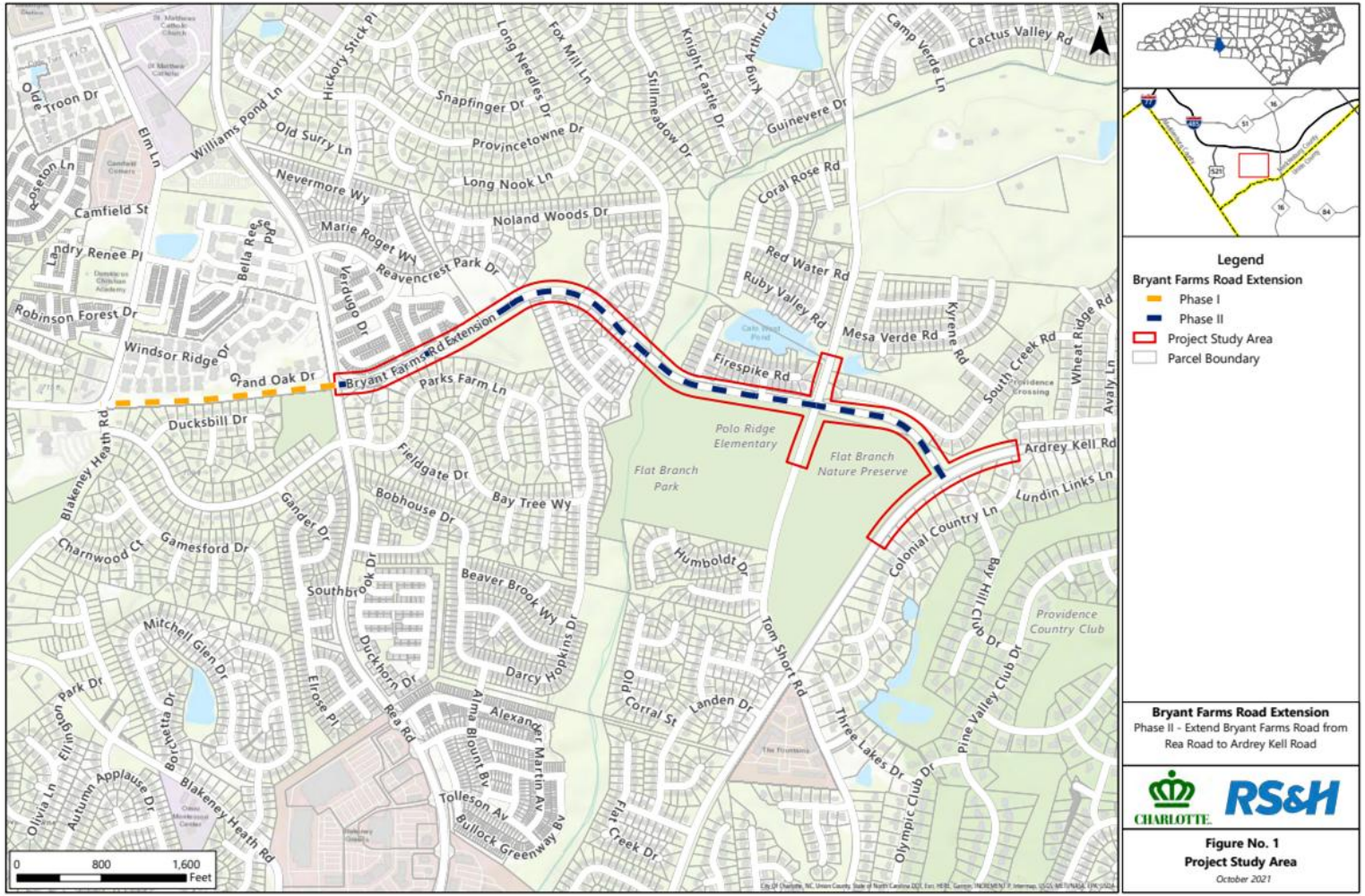


Figure 2: Bryant Farms Road Extension Overall Exhibit (CDOT Bryant Farms RFP 2025)

Problem Statement

The Charlotte Department of Transportation must choose between **hot mix asphalt (HMA)** and **Portland Composite Cement (PCC)** when constructing the Bryant Farms Extension II.

The pavement type selected can significantly impact initial construction cost and long-term maintenance costs.

This capital analysis evaluates which pavement alternative provides the lowest life cycle cost for the Bryant Farms Road Extension Phase II project.

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Why This Project Matters

- The South Charlotte community is in great need of infrastructure investment due to its rapid population growth
- With rising construction costs pavement selection is a high impact financial decision affecting both initial construction and long term maintenance costs.
- Agencies like CDOT must choose between asphalt and concrete with budget and performance constraints.

Roadway Alternatives

- **HMA** (Asphalt)
  - Traditional alternative for roadway construction
  - Benefits: Affordability, paving efficiency, and availability
  - Costs: Expensive rehabilitation maintenance needed every 5-10 years
- **PCC** (Concrete)
  - Specialized alternative
  - Benefits: Low maintenance costs
  - Costs: Higher upfront material cost and lower paving efficiency

(Rao Et. al., 2013)

Objectives

- Make a recommendation to the CDOT of which pavement alternative to select (HMA VS PCC)
- Estimate initial construction cost using public bid tab unit prices.
- Estimate lifecycle costs including O&M and major repairs
- Evaluate cost uncertainty using scenario-based risk analysis.

Capital Cost Analysis

NPV Analysis (Cost Project)

- 30-year life cycle cost analysis
  - Initial Cost
  - Operations and Maintenance
  - Rehabilitation costs
  - Residual Value

Payoff Table - Scenario Analysis

- Expected costs in varied economic conditions
  - S1: Favorable economy
  - S2: Average economy
  - S3: Unfavorable economy

Benefit to Cost Analysis

- Incremental B/C Analysis utilizing expected cash flows
  - ΔO&M
  - ΔCR

Data Sources

Public Bid Tabs

- Publicly available bid reports
- Recent local data for construction costs

Research Articles

- Lifecycle breakdown of pavement alternatives
- Regionally specific research
- O&M economic research

NCDOT Handbook

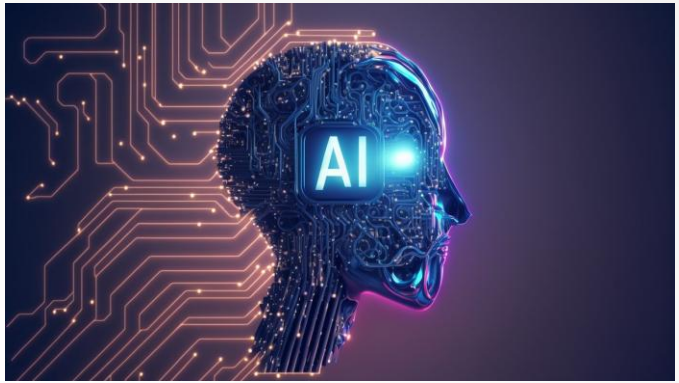
- Design specifications
- Construction Processes
- Maintenance Processes
- General Regulations

Data Standardization

- Necessary process for utilizing real-world data
- Methodology for this will Include
  - Standardize unit costs from data. ie.,
    - $/SY Construction Costs
    - $/SY O&M
    - $/SY Major Repairs
    - $/SY Residual value at end of study period
  - Determine total area (SY) of project pavement
  - Convert monetary data to present value (PV) where necessary

AI Implications

- This project uses LCCA, NPV, and scenario analysis to compare HMA and PCC pavement alternatives.
- AI can strengthen this analysis by improving key inputs such as future material costs and maintenance timing.
- Historical bid tab data can be used to support AI-based forecasting of pavement construction costs.
- Pavement performance data can support AI-based prediction of deterioration and rehabilitation needs.
- AI therefore acts as a supporting tool that improves the accuracy and reliability of traditional capital cost analysis.



Sustainability

- Sustainability factors influence life cycle cost through material usage and maintenance frequency.
- Asphalt pavements require more frequent interventions, increasing cumulative life cycle cost over time.
- Concrete pavements have higher initial cost but lower long-term cost due to reduced maintenance.
- Maintenance activities introduce indirect costs such as traffic delays and fuel consumption.
- Sustainability is incorporated into LCCA as a cost-driving factor through material demand, maintenance cycles, and indirect impacts.

Conclusion

- This project will compare the life-cycle costs of asphalt and concrete pavement alternatives for a representative CDOT roadway segment.
- Real NCDOT bid tabulation data will be used to estimate construction costs and develop a 30-year Life Cycle Cost Analysis (LCCA).
- Capital costing methods such as Net Present Value (NPV) will be applied to evaluate the long-term economic performance of each pavement alternative.
- A scenario-based analysis will incorporate uncertainty in material costs and maintenance timing to support a realistic and data-driven comparison.
- The results will demonstrate how life-cycle cost analysis can support more informed pavement investment decisions for the CDOT.

QUESTIONS?

Sources

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